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SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Artificial Intelligence and Machine Learning Department

LAB REPORT

Offline Multimodal Meeting-Intelligence System

Course: Multimodal Artificial Intelligence Lab (2026-27)

Literature Review for Project

Student Name: AMITKUMAR RACHA

PRN: 24070126501

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DETAILED SCOPUS PAPER ANALYSIS

Core Problem Statement of Project : An offline system that combines long-form meeting audio perform speaker attribution, multimodal alignment, event extraction, decision-lineage reconstruction, and explainable participant-influence analysis.

1. Executive Summary

The selected literature forms a coherent technical foundation for the proposed meeting-intelligence system. The papers mainly address five enabling problems: low-latency speaker spotting/diarization, streaming speaker-attributed ASR, speech enhancement and separation, audio-visual diarization/concurrent-speaker detection, and recognition of overlapping speech.

Across the literature, the most important lesson is that long-form meeting analysis should not rely on a single monolithic model. The strongest engineering direction is a modular but tightly aligned pipeline in which preprocessing, speaker tracking, transcription, multimodal fusion, event extraction, and decision analysis exchange timestamps, confidence values, and speaker representations.

For the proposed project, the literature also exposes a clear research gap: these works largely stop at ‘who spoke when’ or ‘who spoke what when’. They do not provide a complete mechanism for connecting a speaker’s contribution to presentation evidence, tracing how that contribution changes the discussion, reconstructing decision lineage, and computing an explainable influence score.

Therefore, the literature supports positioning the project not as another meeting transcription system, but as a decision- and influence-analysis layer built on top of robust multimodal speech processing.

2. Main Findings at a Glance

Technical area	Key lesson	Best-matching paper(s)	Project use	Remaining gap
Low-latency speaker detection	Optimize diarization for the downstream task, not only DER.	Patino et al. (2018)	Fast speaker initialization and target detection	No semantic/decision reasoning
Streaming SA-ASR	Speaker identity can be attached at token level during streaming recognition.	Kanda et al. (2022)	Speaker-content-time representation	No PPT or influence
Speech enhancement	Small causal models can make local processing practical.	Liang et al. (2023); Lay et al. (2025)	Offline/on-device preprocessing	Enhancement is not identity/causality
Continuous separation	Long sequences and overlap require special architectures.	Li et al. (2022)	Overlap front-end	Separated streams still need attribution
Fast/distilled separation	Teacher-student distillation lets a small model approach a large model's accuracy on real meeting-room audio.	Chen et al. (2021)	Laptop-deployable overlap front-end	No speaker-to-decision linkage
Audio-visual diarization	Visual evidence improves robustness when audio is ambiguous.	Wang et al. (2023); Ishizuka et al. (2009); Eliav & Gannot (2026)	Speaker association and overlap handling	No presentation evidence
Online correction	Streaming diarization benefits from bounded backtracking.	Zhen et al. (2024)	Correct early speaker errors before final scoring	No decision-level correction
Overlap detection/counting	Spatial features and speaker count improve overlap handling.	Cornell et al. (2022)	Gate downstream models and estimate uncertainty	No semantic contribution model
End-to-end multi-talker ASR	Joint modeling can reduce the cost of strict separation→ASR cascades.	Tripathi et al. (2020)	Alternative ASR architecture	Limited real-meeting generalization

3. Overall Literature Synthesis

- The literature strongly validates the importance of low latency, overlap handling, and speaker-attributed transcription for meeting systems.
- Audio-only pipelines remain vulnerable to noise, reverberation, simultaneous speech, short turns, and similar voices.
- Audio-visual fusion provides complementary evidence, especially for active/concurrent speaker detection and speaker-to-person association.
- Streaming systems need mechanisms for maintaining speaker identity over long recordings and correcting early mistakes.
- Teacher-student distillation (Chen et al., 2021) shows that laptop-class hardware can run fast separation models without a large accuracy penalty, directly supporting the offline/local-processing requirement.
- The selected papers do not directly solve presentation-aware evidence grounding or participant influence on decisions.
- The proposed research opportunity is therefore at the integration and reasoning layer: speaker-attributed multimodal events → evidence links → interaction graph → decision lineage → explainable influence.

4. Detailed Analysis of Each Paper

1. Enhanced low-latency speaker spotting using selective cluster enrichment

Authors: Patino, J.; Delgado, H.; Evans, N.

Research problem: Low-latency speaker spotting (rapid detection of a known target speaker) in multi-speaker audio streams.

Core method: Reframes speaker diarization around the downstream speaker-spotting task rather than around minimising the diarization error rate (DER), and introduces a selective cluster enrichment algorithm that uses the target-speaker model to guide which segments the diarization sub-system models, improving cluster purity for the target speaker.

Dataset / setting: Built on the AMI Meeting Corpus, using LLSS-specific protocols (not the standard AMI diarization protocols) designed and released by the authors to support low-latency speaker-spotting evaluation and reproducibility.

Main results: The proposed LLSS system detects target speakers with a 16% equal error rate (EER) after the target speaker has been active for only 15 seconds within the multi-speaker stream.

Strengths: Strong application-oriented design; explicitly optimises the upstream diarization component for the final detection task; demonstrates a useful latency/accuracy trade-off; uses a genuine multi-party meeting corpus (AMI).

Limitations: The target is known-speaker spotting rather than full unconstrained meeting diarization; results are reported on one benchmark family (AMI-derived) and may not generalise to arbitrary institutional meeting rooms or speaker counts.

Direct relevance to the proposed project: Useful for fast speaker initialisation, target-speaker detection, and the broader design principle that intermediate components (diarization) should be optimised for the final meeting-intelligence objective rather than DER alone.

Research gap exposed: Does not model semantic content, PPT evidence, decision lineage, or participant influence.

Critical interpretation

This paper is important because it changes the optimisation target: a diarization sub-system should be designed around the downstream application. For the project, this supports optimising speaker tracking not merely for a low DER, but for reliable speaker-event attribution and later influence analysis.

Dimension	Assessment	Project implication
Technical maturity	Strong / established enabling technology	Use as enabling layer, not sole novelty
Fit to project	Medium–High	Integrate where it improves robustness
Novelty contribution	Enabling rather than sufficient	Novelty should be in cross-modal decision/influence reasoning

2. Streaming Speaker-Attributed ASR with Token-Level Speaker Embeddings

Authors: Kanda, N.; Wu, J.; Wu, Y.; Xiao, X.; Meng, Z.; Wang, X.; Gaur, Y.; Chen, Z.; Li, J.; Yoshioka, T.

Research problem: Streaming joint speech recognition and speaker attribution under overlapping speech.

Core method: Uses token-level Serialized Output Training (t-SOT) together with an encoder-decoder-based speaker embedding extractor. A 't-vector' speaker representation is estimated synchronously for every recognised token, enabling joint speaker identification/diarization and multi-talker transcription in a single streaming pass.

Dataset / setting: LibriSpeechMix and LibriCSS.

Main results: The authors report substantially better accuracy than a prior streaming speaker-attributed ASR baseline, and comparable or in some conditions better results than a state-of-the-art offline speaker-attributed ASR system, while operating in a streaming (low-latency) mode.

Strengths: Very relevant to long-form, low-latency speaker-attributed transcription; token-level speaker representations directly connect individual words with speaker identity, avoiding a hard diarization/ASR boundary.

Limitations: Simulated and benchmark datasets (LibriSpeechMix, LibriCSS) do not fully reproduce confidential institutional meetings; semantic/decision-level analysis is outside the model's scope.

Direct relevance to the proposed project: Provides a strong alternative to a strict diarization-then-ASR cascade. Token-level speaker attribution can give cleaner speaker-content-time units for downstream event and influence analysis.

Research gap exposed: No PPT alignment, evidence verification, decision causality, or influence scoring.

Critical interpretation

This is one of the closest technical matches to the project's audio backbone. Token-level speaker embeddings create a fine-grained unit containing speaker, lexical content, and time — ideal for later event and interaction graphs.

Dimension	Assessment	Project implication
Technical maturity	Strong / established enabling technology	Use as enabling layer, not sole novelty
Fit to project	High	Integrate where it improves robustness
Novelty contribution	Enabling rather than sufficient	Novelty should be in cross-modal decision/influence reasoning

3. Real-Time Diffusion Buffer for Speech Enhancement On A Laptop

Authors: Lay, B.; Makarov, R.; Gerkmann, T.

Research problem: Live, real-time demonstration of diffusion-based single-microphone speech enhancement that removes background noise in reverberant, noisy rooms (e.g. loud babble) on ordinary laptop hardware.

Core method: Packages the authors' 'Diffusion Buffer' framework — a sliding-window/temporal diffusion process that progressively denoises audio frames as they age inside a buffer, assigning more noise to frames closer to the present — as an interactive demo. Conference attendees can toggle the enhancement on/off and adjust the quality/latency trade-off live, experiencing sub-second latency directly.

Dataset / setting: No benchmark corpus is evaluated in this 2-page demo paper itself; it is a live show-and-tell running on conference-room audio (loud babble noise, reverberant single-microphone capture). Formal quantitative evaluation (PESQ/WVMOS/latency numbers) belongs to the companion full paper, 'Diffusion Buffer: Online Diffusion-based Speech Enhancement with Sub-Second Latency' (Lay, Makarov, Gerkmann, INTERSPEECH 2025), which reports algorithmic latencies of roughly 320–960 ms on a laptop-class NVIDIA RTX 4080 GPU with real-time factor below 1.

Main results: As a demonstration paper it reports no independent WER/PESQ table; the underlying Diffusion Buffer method it showcases achieves comparable quality to offline diffusion enhancement at 320–960 ms latency, with a controllable quality/latency dial (later Streaming/Buffer follow-up work pushes this toward tens of milliseconds).

Strengths: Shows that generative (diffusion-based) enhancement can be made interactive and run live on consumer laptop hardware rather than only in offline batch evaluation; makes the latency/quality trade-off tangible.

Limitations: Being a 2-page demonstration paper, it does not itself provide a controlled dataset, ablation, or statistical evaluation; those numbers must be sourced from the companion full Diffusion Buffer paper, and should not be attributed to this demo paper directly.

Direct relevance to the proposed project: Relevant whenever meeting recordings contain noise or reverberation. A lightweight, local, laptop-class enhancement front-end (as demonstrated here) could improve diarization and ASR quality while preserving the offline/privacy requirement of the proposed system.

Research gap exposed: Does not address speaker attribution, multimodal alignment, or influence.

Critical interpretation

The paper's main value is as an implementation and feasibility lesson: generative enhancement can be adapted to run interactively, in real time, on a laptop. For the project, enhancement should be treated as an optional, confidence-aware front-end rather than as the main research contribution — and any numerical claims should be drawn from the companion full paper, not this demo abstract.

Dimension	Assessment	Project implication
Technical maturity	Demonstrated feasible on consumer hardware; formal numbers require the companion paper	Use as enabling layer, not sole novelty
Fit to project	Medium–High	Integrate where it improves robustness
Novelty contribution	Enabling rather than sufficient	Novelty should be in cross-modal decision/influence reasoning

4. Lite-RTSE: Exploring a Cost-Effective Lite DNN Model for Real-Time Speech Enhancement in RTC Scenarios

Authors: Liang, X.; Zhang, L.; Wu, Z.; Xu, R.

Research problem: Efficient real-time speech enhancement for computationally constrained real-time-communication (RTC) hardware.

Core method: Combines low-dimensional LSTM units with a multi-order convolution block and a two-stage complex-spectrum reconstruction scheme (masking followed by residual reconstruction).

Dataset / setting: Not fully specified in the supplied abstract-level record.

Main results: The model contains only 1.56 million parameters and requires 0.55 GMAC/s while achieving denoising performance competitive with state-of-the-art enhancement systems.

Strengths: Excellent engineering fit for laptop/edge deployment; explicitly balances quality, model size, and computation for real-time use.

Limitations: Monaural speech enhancement does not by itself solve overlapping speakers, and may suppress useful speaker information if run too aggressively ahead of diarization.

Direct relevance to the proposed project: Good candidate for a lightweight local preprocessing module before VAD/diarization/ASR, especially since the final system must work without cloud GPUs.

Research gap exposed: No semantic or multimodal reasoning; no evidence or decision modeling.

Critical interpretation

The 1.56M-parameter, 0.55 GMAC/s footprint makes this work especially relevant to the requirement for practical local processing. A similar design philosophy can keep the meeting pipeline usable on ordinary laptops.

Dimension	Assessment	Project implication
Technical maturity	Strong / established enabling technology	Use as enabling layer, not sole novelty
Fit to project	Medium–High	Integrate where it improves robustness
Novelty contribution	Enabling rather than sufficient	Novelty should be in cross-modal decision/influence reasoning

5. SKIM: Skipping Memory LSTM for Low-Latency Real-Time Continuous Speech Separation

Authors: Li, C.; Yang, L.; Wang, W.; Qian, Y.

Research problem: Very-low-latency continuous speech separation for meeting audio.

Core method: Uses a time-domain separator with Graph-PIT permutation-invariant training and introduces SKIM (Skipping Memory LSTM) for efficient long-sequence modelling, targeting the long feature sequences produced by low-latency encoders.

Dataset / setting: Simulated meeting-style evaluation; the abstract identifies meeting-style audio but not a full dataset specification.

Main results: SKIM achieves performance on par with or better than DPRNN while reducing computational cost by 75%. The fastest real-time configuration reports a 17.1 dB SDR improvement with under 1 ms of algorithmic latency in the simulated evaluation.

Strengths: Highly relevant to continuous meeting preprocessing; addresses the computational burden of long sequences while preserving low latency.

Limitations: Separation can introduce artefacts and is not equivalent to speaker-identity tracking; simulated meetings may differ from real institutional recordings.

Direct relevance to the proposed project: Potential front-end for overlap handling when ASR/diarization accuracy drops during simultaneous speech.

Research gap exposed: No direct mapping from separated streams to semantic contributions, slides, or decisions.

Critical interpretation

SKIM demonstrates that very low latency does not necessarily require abandoning long-context modelling. This matters because a 1–2 hour meeting contains long-range speaker and topic dependencies even when the acoustic model itself operates on short windows.

Dimension	Assessment	Project implication
Technical maturity	Strong / established enabling technology	Use as enabling layer, not sole novelty
Fit to project	High	Integrate where it improves robustness
Novelty contribution	Enabling rather than sufficient	Novelty should be in cross-modal decision/influence reasoning

6. The Multimodal Information Based Speech Processing (MISP) 2022 Challenge: Audio-Visual Diarization and Recognition

Authors: Wang, Z.; Wu, S.; Chen, H.; He, M.; Du, J.; Lee, C.-H.; Chen, J.; Watanabe, S.; Siniscalchi, S. M.; Scharenborg, O.; et al.

Research problem: Benchmarking audio-visual speaker diarization and recognition in realistic far-field scenarios.

Core method: Defines Audio-Visual Speaker Diarization (AVSD) and Audio-Visual Diarization+Recognition (AVDR) tasks, combining audio and video to address both 'who spoke when' and 'who spoke what when'. The challenge establishes datasets, tracks, baselines, and evaluation conditions.

Dataset / setting: Chinese far-field, home-TV recording scenarios with 2–6 participants and TV background noise.

Main results: The challenge baselines demonstrate the feasibility of AVDR, while the accompanying analysis identifies difficult conditions including far-field video quality, TV noise, and visually indistinguishable speakers.

Strengths: Strong benchmark orientation; explicitly tests multimodal robustness in realistic noisy environments; establishes 'who spoke what when' as a concrete, evaluable task.

Limitations: The scenario is home-TV rather than PPT-driven professional meetings; it does not evaluate decision influence.

Direct relevance to the proposed project: Supports using visual/audio multimodality to resolve speaker ambiguity and overlap. Its 'who spoke what when' framing is directly relevant to the project's timestamped event representation.

Research gap exposed: The modality missing from the challenge, relative to this project, is presentation evidence and decision semantics.

Critical interpretation

MISP is particularly important because it formalises 'who spoke what when' as a multimodal task. The proposed system extends this idea by asking an additional question: what did that contribution do to the decision process?

Dimension	Assessment	Project implication
Technical maturity	Strong / established enabling technology	Use as enabling layer, not sole novelty
Fit to project	High	Integrate where it improves robustness
Novelty contribution	Enabling rather than sufficient	Novelty should be in cross-modal decision/influence reasoning

7. Audio-Visual Approach for Multimodal Concurrent Speaker Detection

Authors: Eliav, A.; Gannot, S.

Research problem: Audio-visual detection of multiple simultaneously active speakers.

Core method: Early fusion of audio and visual features through cross-modal attention with a learnable [CLS] token, evaluated as a multimodal concurrent speaker detection (CSD) system.

Dataset / setting: AMI and EasyCom real-world datasets.

Main results: Experiments support the effectiveness of multimodal fusion; the work reports CSD results on EasyCom, a difficult real-world egocentric-conversation benchmark.

Strengths: Directly addresses overlap, a major failure mode for diarization and ASR; multimodal fusion can exploit visual evidence when acoustics alone are ambiguous.

Limitations: Concurrent speaker detection identifies how many/which candidates are active, not what was said or whether a contribution changed a decision.

Direct relevance to the proposed project: Useful as an overlap detector positioned before speaker attribution and event extraction; visual cues can improve reliability during simultaneous speech.

Research gap exposed: Needs integration with speaker identity, semantic event extraction, PPT evidence, and decision lineage.

Critical interpretation

Concurrent-speaker detection should be conceptually separated from speaker identity. Detecting overlap first can tell the system when speaker attribution is uncertain and when it should invoke stronger multimodal or separation models.

Dimension	Assessment	Project implication
Technical maturity	Strong / established enabling technology	Use as enabling layer, not sole novelty
Fit to project	High	Integrate where it improves robustness
Novelty contribution	Enabling rather than sufficient	Novelty should be in cross-modal decision/influence reasoning

8. A speaker diarization method based on the probabilistic fusion of audio-visual location information

Authors: Ishizuka, K.; Araki, S.; Otsuka, K.; Fujimoto, M.

Research problem: Online audio-visual speaker diarization using physical location evidence.

Core method: Combines speech-presence probabilities from a compact microphone array with face locations from two cameras using probabilistic fusion and online classification.

Dataset / setting: Real casual multi-party conversations with substantial overlap and short speech segments.

Main results: The authors report advantages over conventional (audio-only) diarization approaches, especially under overlap and short-turn conditions.

Strengths: Early demonstration that visual location can stabilise online speaker assignment; explicitly designed for online processing rather than full-recording hindsight.

Limitations: Older feature/model technology; location-based identity is weaker when people are occluded, off-camera, or move substantially.

Direct relevance to the proposed project: Provides historical foundation for audio-visual speaker-to-person association and supports using physical/visual context as an additional confidence signal.

Research gap exposed: No semantic contribution modeling or presentation/decision evidence.

Critical interpretation

This paper provides an early probabilistic foundation for combining physical visual location with audio evidence. Its historical importance is that multimodal speaker assignment is not new; the project must therefore focus its novelty on higher-level evidence and decision reasoning.

Dimension	Assessment	Project implication
Technical maturity	Strong / established enabling technology	Use as enabling layer, not sole novelty
Fit to project	Medium–High	Integrate where it improves robustness
Novelty contribution	Enabling rather than sufficient	Novelty should be in cross-modal decision/influence reasoning

9. Winner Takes It All: An Efficient Overlap-Aware Hybrid Online Diarization with Partial Backtracking Mechanism

Authors: Zhen, R.; Zhang, X.; Min, C.; Li, B.

Research problem: Efficient overlap-aware online diarization with limited correction of early streaming errors.

Core method: Combines incremental clustering and local diarization, adds a transformer segmentation model, and introduces partial backtracking to repair temporary early errors without fully revising all past outputs.

Dataset / setting: AMI, VoxConverse (VOX), and DIHARD III.

Main results: Relative DER decreases of 13.2%, 15.5%, and 9.7% are reported on the three evaluation settings respectively, with over 30% reduction in inference time; a two-pass clustering strategy is shown to further improve offline results.

Strengths: Very strong match for long meetings: streaming systems inevitably make early mistakes, and partial backtracking offers a practical compromise between stability and correction.

Limitations: Backtracking increases system complexity; diarization correction still does not guarantee correct semantic or causal interpretation.

Direct relevance to the proposed project: Highly relevant to long-form meeting processing. The project should consider bounded backtracking so later evidence can correct speaker assignments before influence scores are finalised.

Research gap exposed: No multimodal presentation evidence or decision-influence model.

Critical interpretation

Partial backtracking is highly relevant to the project's explainability requirement. If later evidence reveals that an early speaker label was wrong, the system should be able to revise dependent event and influence records rather than freezing an early error forever.

Dimension	Assessment	Project implication
Technical maturity	Strong / established enabling technology	Use as enabling layer, not sole novelty
Fit to project	High	Integrate where it improves robustness
Novelty contribution	Enabling rather than sufficient	Novelty should be in cross-modal decision/influence reasoning

10. Overlapped Speech Detection and speaker counting using distant microphone arrays

Authors: Cornell, S.; Omologo, M.; Squartini, S.; Vincent, E.

Research problem: Unified overlapped speech detection (OSD) and speaker counting for distant microphones.

Core method: Frames VAD, OSD, joint VAD+OSD, and speaker counting as related supervised classification problems. Compares TCN and Transformer architectures and explores early/late fusion of acoustic and spatial microphone-array features.

Dataset / setting: AMI, CHiME-6, and a purpose-built multichannel synthetic dataset.

Main results: The Transformer architecture performs best among those tested; neural spatial-localisation features improve over single-channel features; training on speaker counting improves OSD compared with VAD+OSD training alone.

Strengths: Strong real-world relevance for meeting audio; demonstrates that spatial information and task formulation both matter.

Limitations: Speaker counting is not identity attribution; distant-microphone conditions require careful hardware assumptions.

Direct relevance to the proposed project: Can be used as an overlap/counting gate before diarization and ASR, reducing downstream confusion and helping estimate uncertainty.

Research gap exposed: No visual modality, presentation evidence, or semantic/decision reasoning.

Critical interpretation

This paper supports an explicit overlap/counting stage. Overlap is not just an ASR problem: it changes diarization confidence and should therefore propagate uncertainty into later influence scores.

Dimension	Assessment	Project implication
Technical maturity	Strong / established enabling technology	Use as enabling layer, not sole novelty
Fit to project	High	Integrate where it improves robustness
Novelty contribution	Enabling rather than sufficient	Novelty should be in cross-modal decision/influence reasoning

11. Ultra fast Speech Separation Model with Teacher Student Learning

Authors: Chen, S.; Wu, Y.; Chen, Z.; Wu, J.; Yoshioka, T.; Liu, S.; Li, J.; Yu, X.

Research problem: Making Transformer-based continuous speech separation fast enough for practical deployment on edge devices without losing the accuracy of large Transformer/Conformer separators.

Core method: Trains a small 'student' Transformer separator (fewer encoder layers) using layer-wise Teacher–Student (T-S) learning against a larger Conformer 'teacher' model, combined with an objective-shifting mechanism so the student learns from intermediate teacher representations as well as the final target. Large amounts of unlabeled speech are exploited during distillation.

Dataset / setting: Training: roughly 2,000 hours of LibriVox-derived mixtures plus 600 hours of in-house (noisier, no clean reference) mixtures, created by mixing single-speaker utterances. Evaluation: LibriCSS — 10 hours of concatenated and mixed LibriSpeech utterances played back and re-recorded in a real meeting room — tested in both single-channel and seven-channel settings, with both utterance-wise and continuous-input evaluation protocols.

Main results: Compared with a same-size Transformer-small model trained from scratch, the proposed T-S learning method reduces word error rate (WER) by more than 5% relative for both single-channel and multi-channel separation on LibriCSS. Exploiting additional unlabeled data for distillation yields a further 10%+ relative WER improvement, allowing the small/fast student model to approach the larger Conformer teacher's accuracy at a fraction of its inference cost.

Strengths: Directly targets the deployment gap between large/slow separators and small/fast ones; evaluated on a realistic meeting-room dataset (LibriCSS) rather than pure simulation; gives a clean latency/accuracy trade-off story achieved through model-size reduction plus distillation rather than only architectural tricks.

Limitations: Still a separation-only front-end that must feed a downstream ASR system; LibriCSS, although recorded in a real room, replays concatenated/simulated LibriSpeech content rather than genuinely spontaneous multi-party meeting speech; no semantic, evidence, or decision-level reasoning is attempted.

Direct relevance to the proposed project: Confirms that fast, laptop-deployable separation front-ends are achievable via teacher-student distillation without major accuracy loss — directly supporting the project's local/offline-processing requirement, and a good candidate to front the diarization+ASR stage specifically for overlap-heavy segments.

Research gap exposed: No speaker-content-to-decision linkage, no PPT alignment, and no influence modelling.

Critical interpretation

With full-text details now confirmed (via the ISCA Archive and the authors' arXiv preprint), this paper is one of the more directly transferable engineering results for a laptop-based, offline separation front-end: the real meeting-room LibriCSS benchmark and the 10%+ relative WER gain from distillation give a concrete, load-bearing data point for the project's overlap-handling layer.

Dimension	Assessment	Project implication
Technical maturity	Strong / established — verified via ISCA Archive	Use as enabling layer, not sole novelty
Fit to project	Medium–High	Integrate where it improves robustness
Novelty contribution	Enabling rather than sufficient	Novelty should be in cross-modal decision/influence reasoning

12. End-To-End Multi-Talker Overlapping Speech Recognition

Authors: Tripathi, A.; Lu, H.; Sak, H.

Research problem: End-to-end recognition of overlapping multi-speaker speech.

Core method: Augments an RNN-Transducer (RNN-T) with a masking model for separation of encoded audio features, multiple label encoders, a masking L2 loss, and a speaker-embedding loss to facilitate speaker tracking.

Dataset / setting: Simulated overlapping speech for training and real-life telephone conversations for evaluation.

Main results: The proposed system reports 32% WER on words in overlapping speech segments; a manual human-transcription comparison on the same test set reports 37% WER, illustrating the intrinsic difficulty of the task.

Strengths: Demonstrates that multi-talker recognition can be learned jointly rather than requiring a strict separation-then-ASR pipeline; includes explicit speaker-tracking supervision.

Limitations: Telephone speech differs from distant meeting audio; the reported WER is high and overlap remains fundamentally difficult.

Direct relevance to the proposed project: Supports exploring joint multi-talker ASR or speaker-attributed ASR instead of assuming perfect separation first.

Research gap exposed: Does not address multimodal evidence, slide alignment, decisions, or influence.

Critical interpretation

This paper demonstrates that multi-talker recognition can be modelled jointly with speaker tracking. For the project, it supports evaluating both cascaded and joint architectures rather than assuming that one pipeline is universally superior.

Dimension	Assessment	Project implication
Technical maturity	Strong / established enabling technology	Use as enabling layer, not sole novelty
Fit to project	Medium–High	Integrate where it improves robustness
Novelty contribution	Enabling rather than sufficient	Novelty should be in cross-modal decision/influence reasoning

5. Comparative Analysis of All Papers

Paper	Main task	Streaming	Overlap	Audio-visual	Speaker-content link	Direct project value
1	Speaker spotting	Yes	Indirect	No	Partial	High
2	SA-ASR	Yes	Yes	No	Strong / token-level	Very High
3	Speech enhancement (demo)	Yes	Not primary	No	No	Medium
4	Speech enhancement	Yes	Not primary	No	No	High for edge deployment
5	Continuous separation	Yes	Yes	No	No	High
6	AV diarization/recognition	Challenge-oriented	Yes	Yes	Strong	Very High
7	Concurrent speaker detection	Yes / real-time oriented	Core	Yes	No	High
8	AV diarization	Online	Yes	Yes	Partial	High
9	Online diarization	Yes	Core	No	Partial	Very High
10	OSD + counting	Yes / low-latency relevant	Core	Spatial audio	No	High
11	Fast/distilled separation	Yes	Core	Not primary	No (speaker-agnostic separation)	Medium–High
12	End-to-end MTASR	Streaming-capable	Core	No	Speaker tracking	High

6. Research Gap and Proposed Novelty

Capability	Covered by selected literature?	Project extension	Research importance
Who spoke when?	Yes, strongly	Use robust diarization/SA-ASR foundation	Necessary but not novel by itself
Who spoke what when?	Yes	Use token/segment speaker attribution	Core input representation
Who was speaking simultaneously?	Yes	Use OSD/CSD + multimodal cues	Important uncertainty signal
What slide/evidence was being discussed?	Not directly	Timestamp + semantic alignment with PPT	Key multimodal extension
Did a spoken claim support/contradict PPT evidence?	Not directly	Claim-evidence matching and contradiction detection	Strong project-specific contribution
How did contributions change the discussion?	Not directly	Interaction/event graph	Core reasoning layer
Which events led to the final decision?	Not directly	Decision lineage graph	Core decision-intelligence contribution
Who influenced the decision?	Not directly in this set	Influence propagation + human validation	Main research question
Can influence be explained?	Not directly	Evidence-backed score with timestamps and source events	Critical for trust and academic evaluation

7. Recommended System Architecture Based on the Literature

Layer	Recommended design	Literature / rationale	Implementation note
Layer 1 — Audio preprocessing	VAD + optional lightweight enhancement	Liang et al.; Lay et al.	Keep enhancement optional and confidence-aware.
Layer 2 — Overlap detection	OSD/CSD + speaker counting	Cornell et al.; Eliav & Gannot	Flag difficult regions and control downstream processing.
Layer 2b — Fast separation (optional)	Distilled/teacher-student separator for overlap-heavy segments	Li et al. (SkiM); Chen et al. (T-S learning)	Reserve for segments the overlap gate flags as difficult; keep laptop-deployable.
Layer 3 — Speaker attribution	Online diarization / SA-ASR / embeddings	Patino et al.; Kanda et al.; Zhen et al.	Prefer timestamped speaker-content units.
Layer 4 — Multimodal association	Audio + visual evidence where video exists	MISP; Ishizuka et al.; Eliav & Gannot	Use visual cues as complementary evidence, not mandatory input.
Layer 5 — PPT understanding	Text/table/OCR/vision extraction	Project-specific	Create slide evidence objects with slide IDs and embeddings.
Layer 6 — Cross-modal alignment	Timestamp + semantic similarity + explicit slide references	Project-specific	Align utterances to candidate slides with confidence.
Layer 7 — Event extraction	Proposal, objection, evidence, correction, agreement, decision, action	Project-specific	Convert transcript into structured events.
Layer 8 — Interaction graph	Speaker-to-speaker response/objection/support edges	Project-specific	Model discussion dynamics.
Layer 9 — Decision lineage	Contribution → evidence → modification → decision	Project-specific	Trace how decisions formed.
Layer 10 — Influence model	Participation + interaction + evidence + decision impact + temporal propagation	Project-specific	Validate against human annotations; avoid equating talk time with influence.

8. Proposed Research Methodology

A defensible experimental methodology should compare the proposed multimodal influence model against increasingly strong baselines.

Baseline	Definition	Purpose
B1 — Talk time	Influence = speaking duration	Tests whether the proposed method beats a naive participation metric.
B2 — Participation	Talk time + number of turns	Adds basic conversational activity.
B3 — Interaction	Participation + responses/objections/agreements	Captures network-level interaction.
B4 — Audio-only semantic	Speaker attribution + event extraction	Tests the value of presentation evidence.
B5 — Audio + PPT	Speaker events + slide alignment + evidence links	Tests multimodal evidence.
B6 — Proposed	Audio + PPT + interaction graph + decision lineage + temporal influence propagation	Full research model.

9. Evaluation Plan

Component	Metrics	Why
ASR	WER / CER	Measures transcription quality.
Diarization	DER / JER / speaker-attributed error	Measures who-spoke-when accuracy.
Overlap	OSD/CSD precision, recall, F1	Measures simultaneous speech detection.
Event extraction	Precision, recall, F1	Measures proposal/objection/evidence/decision/action extraction.
PPT alignment	Precision, recall, F1 / retrieval@K	Measures speech-to-slide evidence association.
Decision extraction	Precision, recall, F1	Measures final decision identification.
Influence ranking	Spearman, Kendall tau, NDCG, Precision@K	Compares model ranking with human judgments.
Explainability	Human rating of evidence sufficiency	Tests whether score explanations are understandable and traceable.
Efficiency	RTF, memory, latency, CPU/GPU utilization	Tests offline laptop feasibility.

10. Human Annotation Protocol for Influence

Influence cannot be validated using WER or DER alone. A separate human-annotation layer is required. Annotators should examine each decision and identify which contributions materially affected its trajectory.

- For every final decision, annotate the initial proposal, objections, supporting evidence, corrections/modifications, final acceptance, and action assignment.
- Assign each relevant contribution an influence category such as negligible, supporting, substantial, or critical.
- Record whether the contribution introduced new information, changed another speaker's position, resolved an objection, or directly affected the final decision.
- Use at least two independent annotators and calculate inter-annotator agreement before using the labels as ground truth.
- Treat model-derived counterfactual influence as an analytical estimate, not proof of causal influence.

11. Key Design Principles Derived from the Literature

- Optimize intermediate components for the final application, not only for generic benchmark metrics.
- Preserve timestamps throughout the pipeline; time is the common coordinate system linking audio, speakers, slides, events, and decisions.
- Represent uncertainty explicitly, especially around overlap, low SNR, missing faces, and early streaming diarization errors.
- Use bounded backtracking so later information can repair early speaker assignments without recomputing the entire meeting.
- Prefer lightweight/causal models for local deployment (including distilled separators, per Chen et al.), and reserve expensive models for difficult segments.
- Do not treat speaking time as influence; use interaction, evidence, and decision impact.
- Every high-level score should have a traceable evidence path back to transcript segments and, when relevant, slide content.
- Evaluate cascaded and joint architectures experimentally rather than assuming one is always superior.

12. Limitations of This Literature Review

- Official Scopus record links for all 12 papers, exactly as supplied for the assignment, are consolidated in Section 15; these should still be opened and cross-checked against the institution's Scopus subscription immediately before formal submission, since link tokens can be session- or account-specific.
- Most available evidence here is based on abstracts, official conference/journal archives (IEEE Xplore, ISCA Archive, ACM DL), and author preprints rather than the complete typeset Scopus abstract-and-citation record.
- Paper 41 (Chen et al., 2021) previously lacked a matching bibliographic record; it has now been fully verified against the ISCA Archive proceedings entry and the authors' arXiv preprint (arXiv:2204.12777), and the analysis in Section 4 has been completed accordingly.
- Paper 24 (Lay et al., 2025) is a 2-page INTERSPEECH Show & Tell (live demonstration) paper rather than a full experimental study; its own record carries no independent quantitative table, so numerical latency figures are explicitly sourced from the companion full paper and clearly labelled as such rather than attributed to the demo paper.
- The Scopus publication-ID string supplied for Paper 9 (Patino et al., 2018) contained characters that render ambiguously in the source list; the ID has been reproduced as supplied, but the DOI (10.23919/BIOSIG.2018.8553619) and IEEE Xplore record should be used as the authoritative cross-check.
- Several benchmark datasets used across these papers are simulated or domain-specific (e.g. LibriCSS, LibriSpeechMix, home-TV MISP recordings), so their reported numbers should not be assumed to transfer directly to university or enterprise meetings.

13. Conclusion

The literature strongly supports the technical feasibility of the project's lower layers: real-time/low-latency enhancement, overlap detection, continuous and distilled separation, online diarization, speaker-attributed ASR, and audio-visual speaker association are all active and technically mature research areas.

The stronger research opportunity is the layer above these components. The selected papers largely stop at identifying speakers, transcribing their content, or improving robustness under overlap. The proposed system can extend this foundation by aligning speaker-attributed speech with PPT evidence, extracting structured meeting events, constructing interaction and decision graphs, and estimating participant influence using evidence-backed temporal propagation.

Therefore, the recommended thesis/project contribution is: an offline, explainable, multimodal meeting-intelligence framework that goes beyond transcription to reconstruct how evidence and participant contributions propagate into decisions. The influence score should be validated against human judgments and presented as an evidence-grounded analytical measure rather than an absolute causal truth.

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